INCH-POUND
MIL-DTL-23053/11C
17 June 1996
SUPERSEDING
MIL-I-23053/11B
01 May 1986

# DETAIL SPECIFICATION SHEET

## INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, FLUORINATED ETHYLENE PROPYLENE, NON-CROSSLINKED

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

PolYmer type: The base polymer used in formulating this sleeving shall be fluorinated ethylene propylene.

Continuous operating temperature range: -67°C (-88°F) to +200°C (+392°F).

Color: The sleeving shall be furnished in clear (water white to transparent blue), the as manufactured color.

Class: The sleeving shall be furnished in the following classes, as specified (see 6.2a):

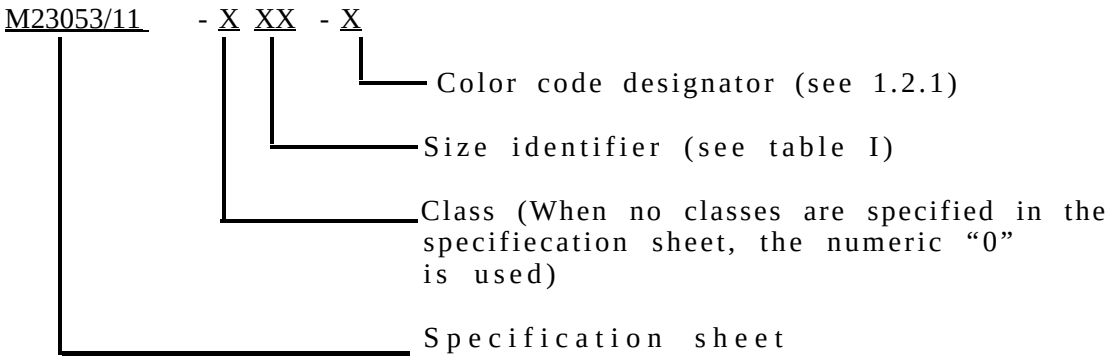
Class 1 - Normal shrink ratio, Heavy Wall
Class 2 - High shrink ratio, Medium Wall

Longitudinal change: ± 15 percent

Military part number: The Military part number shall consist of this specification sheet and dash numbers as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:20Z
Check the source to verify that this is the current version before use.




Example: Class 2, 0.750 inch (19.1 mm) as supplied ID sleeving shall be identified as M23053/11-207-C.

TABLE I. Construction details, inches (mm). 1/

| Military part number | As supplied ID minimum | After unrestristed shrinkage | |
|---|---|---|---|
| | | ID maximum | Wall thickness 2/ |
| Class 1 | | | |
| M23053/11-101-C | .031(.79) | .027(.68) | .008 ± .002(.20 ± .05) |
| M23053/11-102-C | .036(.91) | .032(.80) | .008 ± .002(.20 ± .05) |
| M23053/11-103-C | .045(1.1) | .039(1.0) | .008 ± .002(.20 ± .05) |
| M23053/11-104-C | .060(1.5) | .049(1.2) | .008 ± .002(.20 ± .05) |
| M23053/11-105-C | .075(1.9) | .061(1.6) | .009 ± .002(.23 ± .05) |
| M23053/11-106-C | .092(2.3) | .072(1.8) | .009 ± .002(.23 ± .05) |
| M23053/11-107-C | .115(2.9) | .089(2.3) | .009 ± .002(.23 ± .05) |
| M23053/11-108-C | .141(3.6) | .114(2.9) | .010 ± .003(.25 ± .08) |
| M23053/11-109-C | .158(4.0) | .124(3.2) | .010 ± .003(.25 ± .08) |
| M23053/11-110-C | .180(4.6) | .143(3.6) | .010 ± .003(.25 ± .08) |
| M23053/11-111-C | .197(5.0) | .158(4.0) | .011 ± .004(.28 ± .10) |
| M23053/11-112-C | .225(5.7) | .180(4.6) | .011 ± .004(.28 ± .10) |
| M23053/11-113-C | .248(6.4) | .198(5.0) | .011 ± .004(.28 ± .10) |
| M23053/11-114-C | .290(7.4) | .226(5.8) | .011 ± .004(.28 ± .10) |
| M23053/11-115-C | .310(7.9) | .249(6.4) | .011 ± .004(.28 ± .10) |
| M23053/11-116-C | .365(9.2) | .280(7.2) | .012 ± .004(.30 ± .10) |
| M23053/11-117-C | .400(10.2) | .311(7.9) | .012 ± .004(.30 ± .10) |
| M23053/11-118-C | .440(11.2) | .349(8.9) | .012 ± .004(.30 ± .10) |
| M23053/11-119-C | .500(12.7) | .383(9.7) | .015 ± .004(.38 ± .10) |
| M23053/11-120-C | .580(14.7) | .448(11.4) | .020 ± .004(.50 ± .10) |
| M23053/11-121-C | .666(16.9) | .510(13.0) | .020 ± .004(.50 ± .10) |



Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:20Z
Check the source to verify that this is the currentversion before use.

TABLE I. Construction details, inches (mm). 1/ - Continued

| Military part number | As supplied ID minimum | After unrestricted shrinkage ID maximum | After unrestricted shrinkage Wall thickness 2/ |
|---|---|---|---|
| M23053/11-122-C | .830(21.1) | .637(16.1) | .025 ± .004(.64 ± .10) |
| M23053/11-123-C | 1.000(25.4) | .764(18.9) | .030 ± .004(.80 ± .10) |
| M23053/11-124-C | 1.170(29.7) | .891(22.7) | .035 ± .004(.90 ± .10) |
| M23053/11-125-C | 1.300(33.0) | 1.000(25.4) | .020 ± .004(.50 ± .10) |
| M23053/11-126-C | 1.330(33.8) | 1.020(25.9) | .035 ± .004(.90 ± .10) |
| M23053/11-127-C | 1.700(43.2) | 1.300(33.4) | .020 ± .004(.50 ± .10) |
| M23053/11-128-C | 2.100(53.3) | 1.700(43.2) | .020 ± .004(.50 ± .10) |
| M23053/11-129-C | 2.600(66.0) | 2.100(53.5) | .020 ± .004(.50 ± .10) |
| M23053/11-130-C | 3.100(78.7) | 2.600(66.0) | .020 ± .004(.50 ± .10) |
| M23053/11-131-C | 3.500(88.9) | 3.100(78.7) | .020 ± .004(.50 ± .10) |
| M23053/11-132-C | 4.300(109.2) | 3.500(88.9) | .020 ± .004(.50 ± .10) |
| M23053/11-133-C | 1.500(38.1) | 1.145(29.1) | .035 ± .004(.90 ± .10) |
| M23053/11-134-C | 1.666(42.3) | 1.270(32.3) | .035 ± .004(.90 ± .10) |
| M23053/11-135-C | 1.833(46.6) | 1.390(35.3) | .035 ± .004(.90 ± .10) |
| M23053/11-136-C | 2.000(50.8) | 1.570(38.6) | .035 ± .004(.90 ± .10) |
| Class 2 | | | |
| M23053/11-201-C | .093(2.4) | .056(1.4) | .008 ± .003(.20 ± .08) |
| M23053/11-202-C | .125(3.2) | .075(1.9) | .010 ± .003(.25 ± .08) |
| M23053/11-203-C | .188(4.8) | .115(2.9) | .010 ±.003(.25 ± .08) |
| M23053/11-204-C | .250(6.4) | .150(3.8) | .010 ± .003(.25 ± .08) |
| M23053/11-205-C | .375(9.5) | .225(5.7) | .012 ± .003(.30 ±.08) |
| M23053/11-206-C | .500(12.7) | .300(7.6) | .015 ± .004(.38 ± .10) |
| M23053/11-207-C | .750(19.1) | .450(11.3) | .020 ±.004(.50 ± .10) |
| M23053/11-208-C | 1.000(25.4) | .600(15.0) | .025 ± .005(.64 ± .12) |
| M23053/11-209-C | 1.500(38.0) | .900(22.5) | .030 ± .005(.80 ± .12) |
| M23053/11-210-C | 2.000(51.0) | 1.200(30.0) | .030 ± .005(.80 ± .12) |

1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5: Sleeving 1 inch and below: 210° ± 6°C (410° ± 10°F). Sleeving over 1 inch dia. 221° +6°C (430° ± 10°F).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:20Z
Check the source to verify that this is the current version before use.

TABLE II. Physical properties. 1/

| Characteristic | Requirement | Test Procedure and conditions |
|---|---|---|
| As supplied: | | |
| ID, minimum | Table I | 4.6.3 |
| Restricted shrinkage | No cracks | 4.6.6.1.1 175° ± 2°C (347° ± 4°F) |
| Voltage withstand | Pass | 4.6.6.2 |
| Clarity stability | Pass | 4.6.16 175° ± 1°C (347° ± 2°F), 24 hours |
| After unrestricted shrinkage: | | |
| ID, maximum | Table I | 4.6.3 |
| Wall thickness | Table I | 4.6.3 |
| Tensile strength psi (MPa), minimum | 2,000 (13.8) | 4.6.13 ASTM D638, 2 inches/minute |
| Ultimate elongation, percent, minimum | 200 | 4.6.13 ASTM D638, 2 inches/minute |
| Dielectric strength, volts/roil (Kv/mm), minimum | 500(12.7) | 4.6.2 ASTM D2671 |
| Volume resistivity, Ohm-cm, minimum | $10^{17}$ | 4.6.2 ASTM D876 |
| Water aborption, percent, maximum | 0.01 | 4.6.2 ASTM D570, 24 hrs at 23°C |
| Low temperature flexibility | No cracks | 4.6.7.1 -65° ± 1°C (-85° ± 2°F) |
| Heat resistance, properties after: | | 4.6.9 250° ± 3°C (482° ± 5.4°F), 96 hours |
| Tensile strength, psi (MPa), minimum | 2000(13.8) | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:20Z
Check the source to verify that this is the current version before use.

TABLE II. Physical properties. 1/ - Continued

| Characteristic | Requirement | Test procedure and conditions |
|---|---|---|
| Ultimate elongation, percent, minimum | 200 | |
| Dielectric strength, volts/roil (Kv/mm), minimum | 500(12.7) | |
| Flammability | Non-flammable | 2/ |
| Corrosion | Non-corrosive | 2/ |
| Fluid resistance | No change | 2/ |
| Fungus resistance | No growth | 2/ |

1/ Unless otherwise specified, the requirements, test procedures and conditions are for all classes.

2/ Because of their relative inertness, MIL-I-23053/11 sleevings are not normally subject to these requirements. However, the acquiring activity (see 6.2k) may, at its option, specify performance of any or all of these requirements.

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 4 years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: Heat shrinkable fluorinated ethylene propylene sleevings are intended for use in high temperature applications.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - MI
Navy - EC, SH
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

(Project 5970-1129-11)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:20Z
Check the source to verify that this is the current version before use.